

Kinabalu Basin

Offshore NW Sabah, Malaysia — a geological dossier

Version 3 · October 2026 · Prepared for Arif Fazil
Public sources only. No proprietary well, seismic or company data.

What changed in v3. The regional section now runs from the Dangerous Grounds to East Sabah and the Sulu Sea (~830 km) and carries four competing working models of the tectonic driver, including a new, untested hypothesis: a basaltic ridge along the Sabah Trough. Errors in v2 are corrected (Section 10).

1. What is the Kinabalu Basin?

In Malaysian petroleum usage the Kinabalu Basin is the offshore sedimentary basin along the north-west Sabah margin, from the waters off Labuan north toward the Malaysia–Philippines boundary. It is one of three producing basins in Malaysia, with the Malay and Sarawak basins (PETRONAS MPM 2025). It spans the NW Sabah shelf (<100 m water) and the deepwater fold-thrust belt (>1,000 m). Its outboard, north-western limit is the **Sabah Trough** (~2.9 km deep), beyond which lie the Layang-Layang Basin and the Dangerous Grounds.

Fields include Kinabalu (~500 MMstb oil in place; Bait 2003), Samarang, Keabangan, Sumandak and Malikai. Recent wells have found oil beneath gas in deeper intervals (Ab Ghani et al. 2026).

2. Tectonic evolution

Sabah sits where the South China Sea, Sulu Sea and Celebes Sea basins meet. Its Cenozoic history runs from subduction, through collision, to post-subduction uplift and gravity-driven deformation.

Age	Event	Record
Eocene–Oligocene	Proto-South China Sea subducts SE beneath NW Borneo	Rajang–Crocker deep-marine turbidites accreted into the prism that forms the Crocker Range (Hutchison 2005)
~37 Ma	Sarawak Orogeny	Uplift and erosion of the Rajang Group (Hutchison 1996)
~32–16 Ma	South China Sea spreading	Dangerous Grounds blocks rifted and drifted south; carbonate platforms on block crests (Brais et al. 1993; Li et al. 2014)
~23–20 Ma	Sabah deformation · BMU	Uplift, erosion, mélange; deep-water to shallower sedimentation (Balaguru & Hall 2009)
~20–15 Ma	Sulu Sea back-arc opening	Extension and subsidence in East Sabah (Rangin & Silver 1991)
~16–15.5 Ma	End of spreading (RU/MMU) · DRU	Platforms drown in the Dangerous Grounds; Deep Regional Unconformity offshore Sabah (Hazebroek & Tan 1993)
Middle Miocene	North Sabah–Pagasa Wedge loaded; mud canopy	Mini-basins and ~1,900 km ² of mud canopy (Morley et al. 2023)
~9.5–8.5 Ma	SRU · Late Miocene inversion	Shallow Regional Unconformity; published ages vary with biostratigraphic calibration
7.85–7.22 Ma	Kinabalu granite	U-Pb zircon (Cottam et al. 2010); rapid exhumation afterwards (Cottam et al. 2013)
Late Miocene–present	Delta progradation, toe-thrusts, inversion	Stage IV deltas; deepwater fold-thrust belt; 2015 Mw 6.0 Ranau earthquake shows ongoing deformation

What drives the deformation is disputed. Section 8 sets out four working models.

3. Stratigraphy

The producing interval is the post-DRU Stage IV succession, subdivided into IVA–IVG (Levell 1987). On the shelf it is a coastal to shallow-marine clastic system thickening into growth faults; outboard it passes into deepwater turbidites and mud, folded above an overpressured shale. Stage IVC records coastal progradation with mouth-bar and channel sands; IVD stacked coarsening-upward shoal complexes; IVE gradual transgression; IVF widespread transgression with laterally extensive sand sheets; IVG renewed shallowing (Bait 2003).

4. Kinabalu Field

Discovered in 1989 by KN-1, 55 km WNW of Labuan in ~54 m of water; oil in place ~500 MMstb; first oil December 1997, peak ~48,000 bbl/d. The trap is a dip closure against the SW–NE Kinabalu growth fault (8° WNW dip; spill to the NE). Main, Deep and East accumulations; best reservoir facies are poorly stratified sandstones (23% porosity, ~630 mD). A water-injection scheme from shallow aquifer sands raised recovery from ~21% to ~43% (Bait 2003).

5. Petroleum system

Element	Summary
Source	Not penetrated. Type III and II/III deltaic organic matter inferred from migrated hydrocarbons; generation from Late Miocene–Pliocene to present.
Reservoir	Post-DRU Stage IV coastal to shallow-marine sandstones (porosity 20–30%); deepwater turbidites outboard; Oligo–Miocene carbonates in the neighbouring Layang-Layang Basin (Tepat-1; Choi 2026).
Seal	Intraformational and transgressive marine shales (IVF, IVG).
Trap	Growth-fault dip closures on the shelf; thrust anticlines outboard; stratigraphic traps under-explored.
Key risk	Trap timing versus charge; Late Miocene–Pliocene inversion can breach or re-migrate.

6. Recent discoveries

Deeper oil beneath gas at Well A (2025), a deeper pool at Zoisit Deep-1, and reprocessed-seismic targets at Well B (Ab Ghani et al. 2026) have shifted the basin from gas-dominant to oil-prone. In the Layang-Layang Basin next door, carbonate discoveries (Tepat-1, Megah-1) prove an Oligo–Miocene carbonate play (Choi 2026). Receiver functions image a slab remnant 45–55 km beneath Sabah (Cornwell et al. 2025).

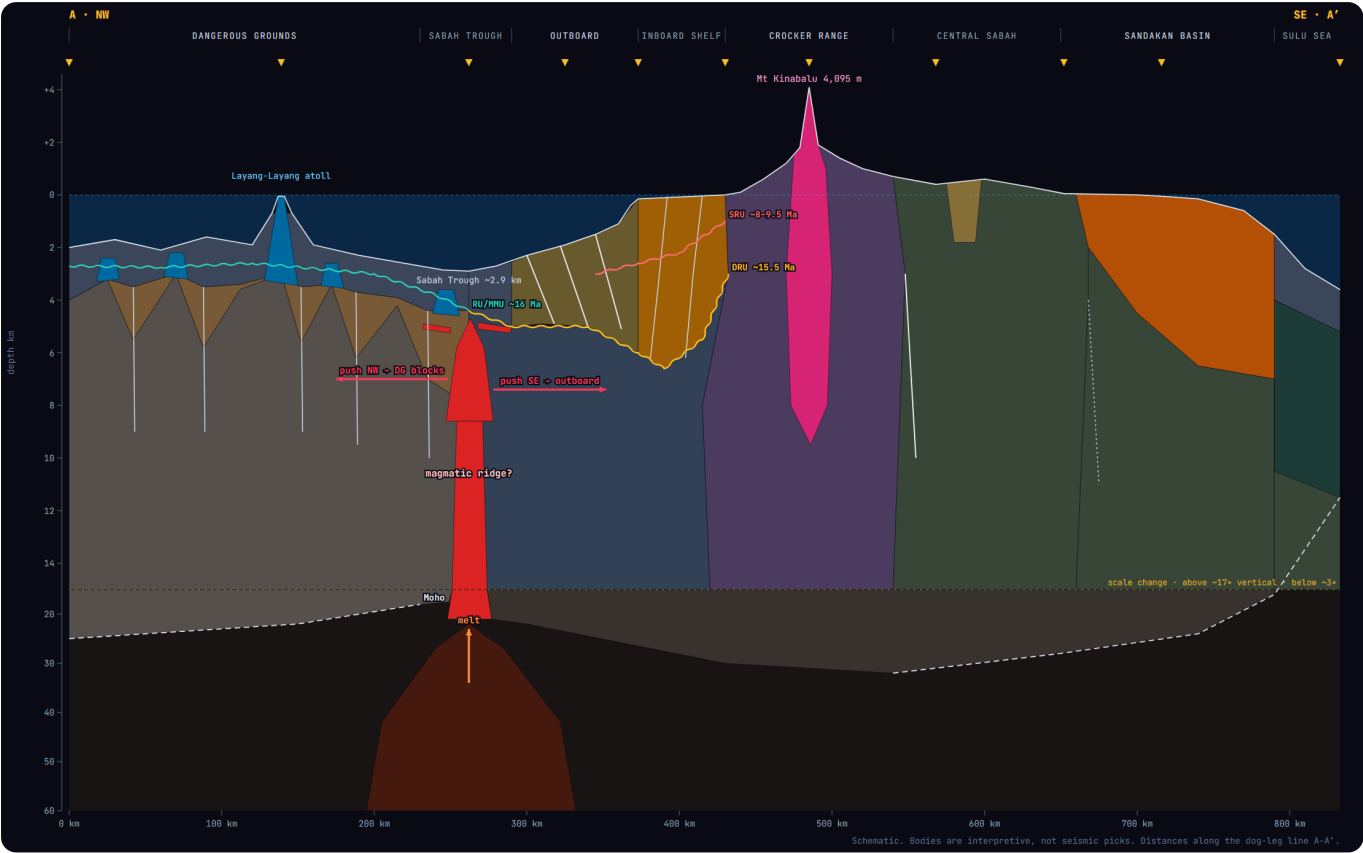
7. Regional section: Dangerous Grounds → East Sabah

An ~830-km dog-leg section from the Dangerous Grounds (NW) across the Sabah Trough, the outboard fold-thrust belt, the Kinabalu shelf, the Crocker Range and Mt Kinabalu, Central Sabah and the Sandakan Basin to the Sulu Sea (SE). The upper 15 km is drawn at ~17× vertical exaggeration and the lithosphere at ~3×. Shallow geometry is shared by all models; the region beneath the DRU and everything below 15 km depend on the model. All bodies are schematic interpretations, not seismic picks. Interactive version: arif-fazil.com/earth/kinabalu-cross-section.html.

8. Four working models

● Model A · new hypothesis — Sabah Trough magmatic ridge UNTESTED

Working hypothesis · Arif Fazil · 2026



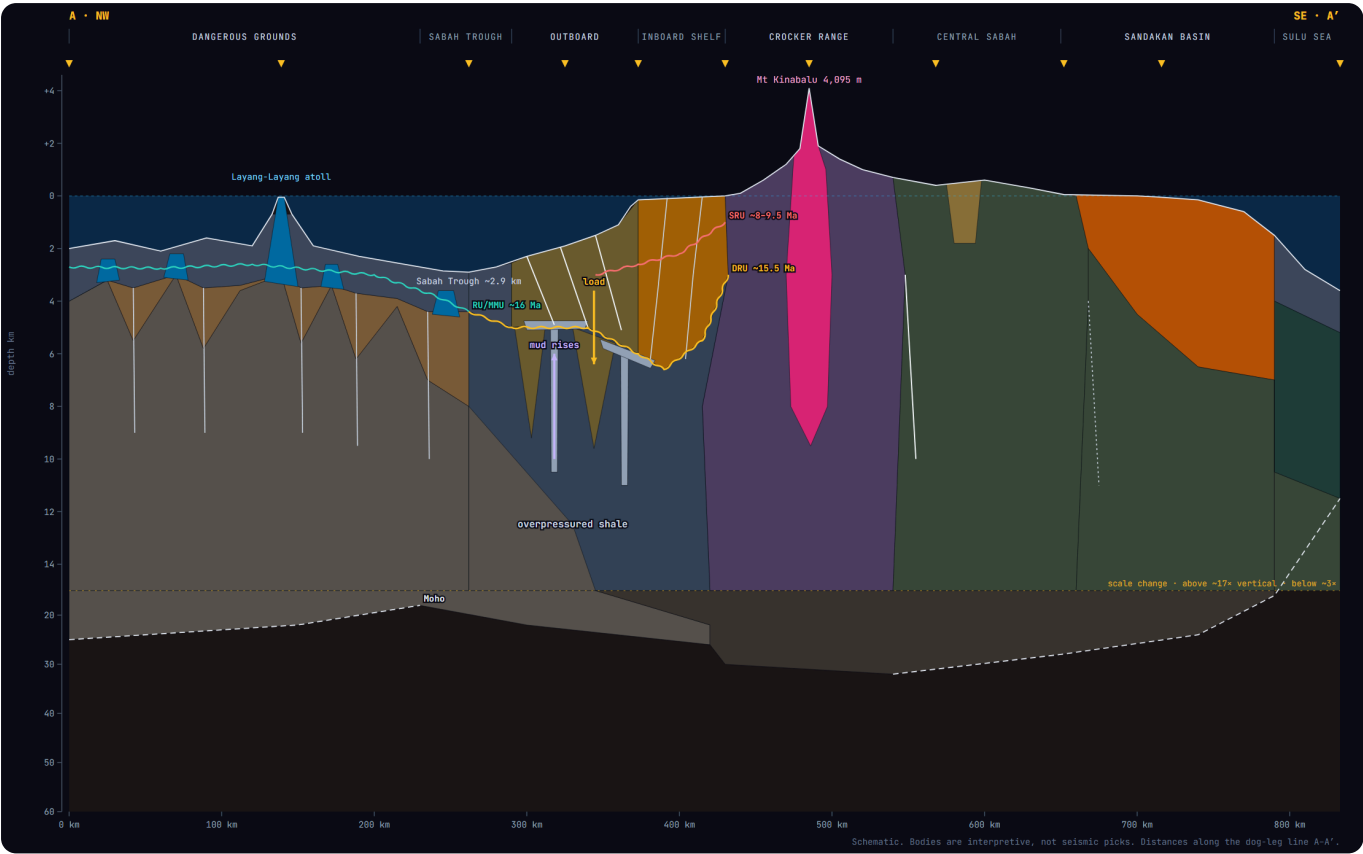
Section A-A' under Model A · new hypothesis.

An extensional basaltic ridge runs along the Sabah Trough, on a strike line through the Tepat-1 area. Magmatic accretion along that axis pushed outward in both directions: north-west into the Dangerous Grounds, reactivating fault blocks and lifting carbonate highs, and south-east into the Kinabalu outboard, adding a driver for toe-thrust and wedge shortening. If real, this is a tectonic event that no current tectonostratigraphic chart records.

MECHANISM	Ridge push / magmatic wedging from an axial dyke complex fed by mantle upwelling beneath the trough.
TIMING	Not yet constrained. Two windows to test: (1) Oligocene–Early Miocene, at the end of rifting; (2) Middle–Late Miocene, coeval with DRU–SRU deformation.
STRENGTHS	Would give the outboard belt an internal driver independent of delta loading; would add a heat source relevant to the new deep oil (oil beneath gas); would tie volcanic or high-standing features along the trough into one structure.
PROBLEMS	No published magnetic lineations, gravity high or dated basalt along the trough axis. The trough is usually read as a flexural foredeep, which predicts a cold, gravity-low axis — the opposite of a ridge. Ridge push is of order 10^{12} N/m even for mature ocean ridges and smaller for a narrow young one; a stress budget is needed to show it can move a 100-km fold-thrust belt.
KILL TEST	No axial magnetic and gravity high along the strike line, and no dated basalt in trough-flank wells → reject.

● **Model B · Morley et al. 2023 — Miocene mud canopy (NSPW)** PUBLISHED

Morley et al. (2023) · Geosphere 19(1): 291 · 3D seismic + wells



Section A-A' under Model B · Morley et al. 2023.

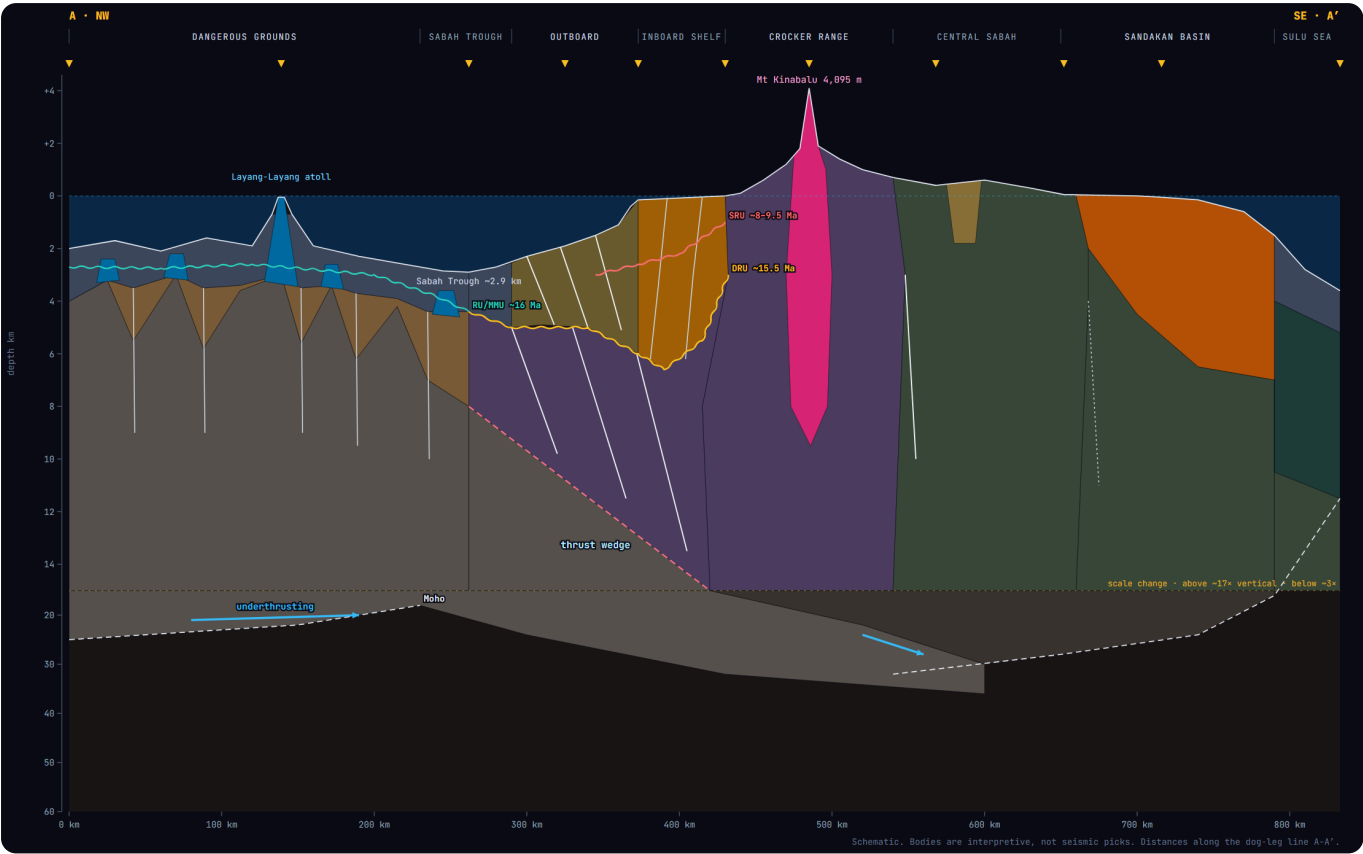
The North Sabah–Pagasa Wedge is not mainly an accretionary wedge. After collision it was a body of overpressured shale, loaded by up to ~4 km of clastics in mini-basins. A short burst of mud volcanism then built a submarine mud canopy: about 50 mud-volcano centres, each 4–80 km², with ~1,900 km² of dense canopy — the largest and most complex such system described.

MECHANISM	Differential sedimentary loading of overpressured shale → shale mobility, mud pipes, chambers and volcanoes.
TIMING	Middle Miocene, after collision.
STRENGTHS	Built on 3D seismic and wells. Explains chaotic seismic facies, mini-basins and vent complexes in the outboard without invoking stacked thrust sheets.
PROBLEMS	Explains the wedge's internal structure, not the plate-scale driver: it still needs one of A, C or D underneath. Canopy level must be reconciled with the DRU age.
KILL TEST	Wells through 'canopy' facies that prove in-situ accreted strata rather than extruded mud → reject for that area.

● **Model C · classic — Continental underthrusting**

CLASSIC

Hinz et al. (1989); Hazebroek & Tan (1993); Hutchison (2005)



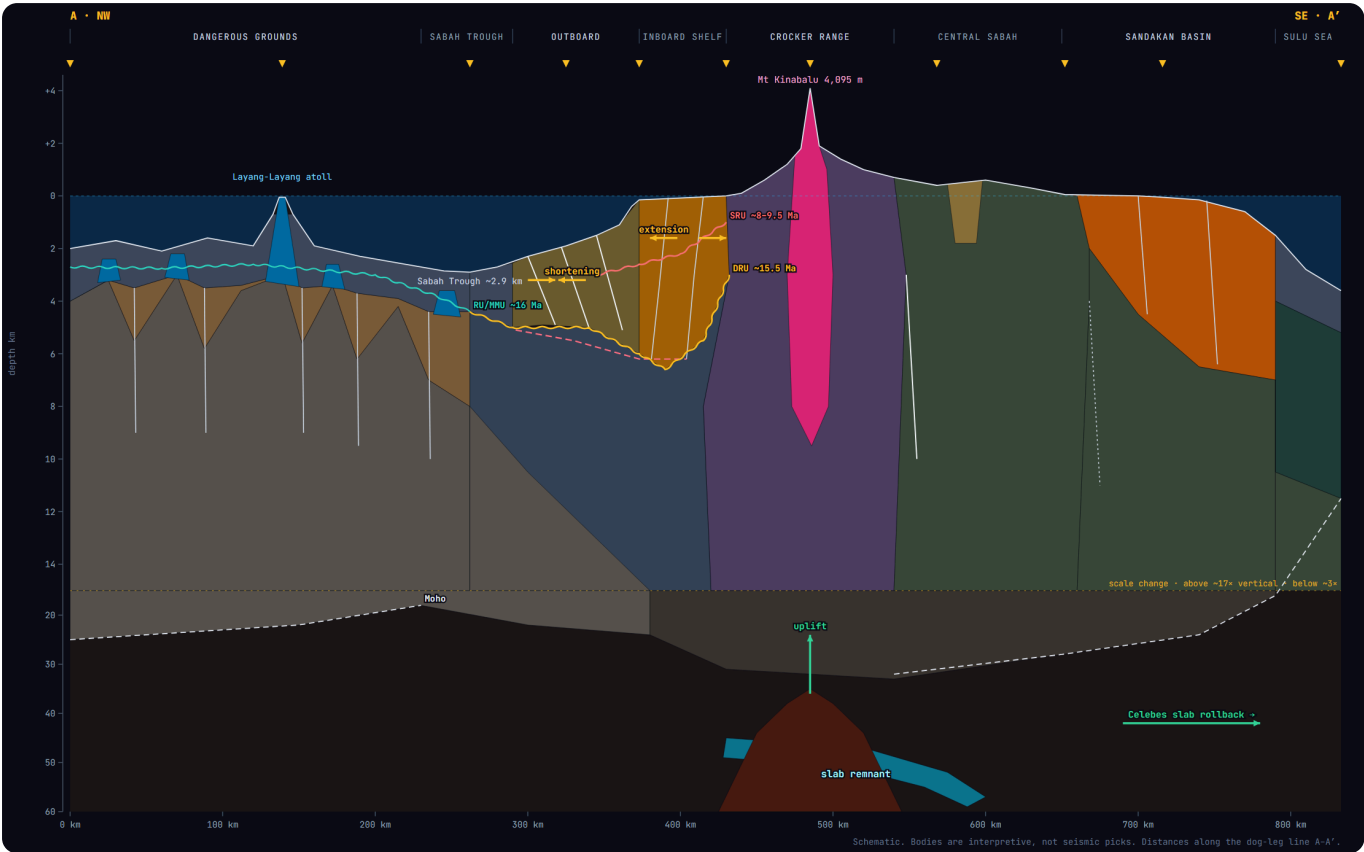
Section A–A’ under Model C · classic.

As the proto-South China Sea closed, Dangerous Grounds continental crust was carried south-east beneath NW Borneo. The Sabah Trough is the flexural foredeep, or remnant trench, on the down-going plate; the North Sabah–Pagasa Wedge is an accretionary thrust wedge; the DRU records the collision.

MECHANISM	Plate convergence and crustal underthrusting; flexural loading of the lower plate.
TIMING	Early to Middle Miocene collision; flexural loading continues.
STRENGTHS	Explains the deep, narrow trough, its flexural normal faults, the imbricate thrusts on older 2D seismic and the regional DRU.
PROBLEMS	Spreading ended ~16 Ma, yet shortening continues to the present; much young shortening is linked to delta loading. 3D seismic shows mobile shale where this model draws thrust sheets (B).
KILL TEST	Deep seismic or receiver functions show no Dangerous Grounds crust beneath the shelf → reject.

● **Model D · post-subduction — Slab remnant + gravity tectonics** PUBLISHED

Hall (2013); Cottam et al. (2010, 2013); King et al. (2010); Cornwell et al. (2025)



Section A–A' under Model D · post-subduction.

Subduction stopped in the Early Miocene, but a detached slab remnant still sits 45–55 km beneath Sabah. Post-subduction mantle flow and rollback of the Celebes Sea slab to the south-east drive uplift (Crocker Range, Kinabalu granite) and extension. The eroding mountains feed deltas whose weight drives shelf extension linked, through a shale detachment, to outboard toe-thrusts.

MECHANISM	Vertical mantle forces + rollback-driven extension + gravitational collapse of the delta wedge.
TIMING	Early Miocene end of subduction; Late Miocene to present uplift, granite and gravity tectonics.
STRENGTHS	Explains the slab image at 45–55 km, granite timing and fast exhumation, and the kinematic link between growth faults and toe-thrusts.
PROBLEMS	Gravity alone may not balance all outboard shortening (King et al. 2010 separate plate-scale and gravitational parts); the timing of Celebes Sea rollback is debated.
KILL TEST	Outboard shortening far exceeds shelf extension with no detachment linkage → a plate-scale push (A or C) is needed.

9. What would decide it

Observation	A	B	C	D	Status
Linear magnetic + gravity high along the trough axis (Tepat-1 strike line)	✓	○	✗	○	open
Dated basalt / dolerite in trough-flank wells	✓	○	○	○	open
Sills, saucer intrusions, hydrothermal vents in trough fill (3D seismic)	✓	○	✗	○	open
Heat flow / maturity peaks on the trough axis	✓	○	✗	○	open
Mud-volcano canopy and pipes at Middle Miocene level	○	✓	✗	○	observed Morley et al. 2023
Outboard shortening exceeds shelf extension	✓	○	✓	✗	open contested; King et al. 2010
Slab remnant at 45–55 km beneath Sabah	○	○	✓	✓	observed Cornwell et al. 2025
Granite 7.85–7.22 Ma with rapid exhumation	○	○	○	✓	observed Cottam et al. 2010, 2013

✓ predicted · ✗ contradicted or not expected · ○ silent. The four observations that would test Model A exist in industry data but are not public. Model A is drawn at the same weight as the published models so it can be argued with, not because it is established.

10. Corrections from v2 (July 2026)

- Kinabalu granite: v2 gave ~10–8 Ma. The U-Pb age is **7.85–7.22 Ma** (Cottam et al. 2010). The unsourced "~1,000 km² at depth" is removed.
- Sabah Trough: v2 called it the basin's eastern boundary. It is the **outboard, north-western** limit.
- SRU: v2 gave a single 8.6 Ma. Published ages range ~8.5–9.5 Ma depending on biostratigraphic calibration; it is shown as a range.
- The regional section is extended from ~600 km to ~830 km to reach the Sulu Sea, and now carries four models instead of one picture.

11. References

1. Fazil, A. (2026). Sabah Trough magmatic-ridge hypothesis. Working note, this page (untested).
2. Morley, C.K. et al. (2023). A major Miocene deepwater mud canopy system: the North Sabah–Pagasa Wedge, northwestern Borneo. *Geosphere* 19(1): 291.
3. Hinz, K. et al. (1989). Thrust tectonics along the north-western continental margin of Sabah/Borneo. *Geologische Rundschau* 78: 705–730.
4. Hazebroek, H.P. & Tan, D.N.K. (1993). Tertiary tectonic evolution of the NW Sabah continental margin. *Bulletin of the Geological Society of Malaysia* 33: 195–210.
5. Hutchison, C.S. (2005). *Geology of North-West Borneo: Sarawak, Brunei and Sabah*. Elsevier.
6. Hall, R. (2013). Contraction and extension in northern Borneo driven by subduction rollback. *Journal of Asian Earth Sciences* 76: 399–411.
7. Cottam, M.A. et al. (2010). Pulsed emplacement of the Mount Kinabalu granite, northern Borneo. *Journal of the Geological Society* 167: 49–60.
8. Cottam, M.A. et al. (2013). Neogene rock uplift and erosion in northern Borneo: evidence from the Kinabalu granite, Mount Kinabalu. *Journal of the Geological Society* 170: 805–816.

9. King, R.C. et al. (2010). Balancing deformation in NW Borneo: quantifying plate-scale vs. gravitational tectonics in a delta and deepwater fold-thrust belt system. *Marine and Petroleum Geology* 27: 238–246.
10. Cornwell et al. (2025). *Journal of Geophysical Research* — teleseismic receiver functions beneath Sabah (as cited in the dossier; full reference to be confirmed).
11. Balaguru, A. & Hall, R. (2009). Tectonic evolution and sedimentation of Sabah, North Borneo. *AAPG Search and Discovery* #30084.
12. Balaguru, A., Nichols, G. & Hall, R. (2003). The origin of the 'circular basins' of Sabah. *Bulletin of the Geological Society of Malaysia* 46: 335–351.
13. Madon, M. & Jong, J. (2022). Structural-stratigraphic framework and petroleum systems of the Sandakan Basin. *Bulletin of the Geological Society of Malaysia* 74: 135–160.
14. Hinz, K. & Schlüter, H.U. (1985). Geology of the Dangerous Grounds, South China Sea, and the continental margin off southwest Palawan. *Energy* 10: 297–315.
15. Biais, A., Patriat, P. & Tapponnier, P. (1993). Updated interpretation of magnetic anomalies and seafloor spreading stages in the South China Sea. *JGR* 98: 6299–6328.
16. Li, C.-F. et al. (2014). Ages and magnetic structures of the South China Sea constrained by deep tow magnetic surveys and IODP Expedition 349. *G³* 15: 4958–4983.
17. Rangin, C. & Silver, E.A. (1991). Neogene tectonic evolution of the Celebes-Sulu basins. *Proceedings of the Ocean Drilling Program, Scientific Results* 124.
18. Choi, T. (2026). Unlocking the Oligocene play in offshore Sabah: new insights from Layang-Layang Basin. TGS / EAGE workshop.
19. Ab Ghani, A.F. et al. (2026). Unravelling deeper play opportunities in Kinabalu Basin of Sabah: case study from recent Well B results. 2nd EAGE Workshop on New Discoveries in Mature Basins.
20. Bait, B. (2003). Geology of Kinabalu field and its water injection scheme. *Bulletin of the Geological Society of Malaysia* 47: 165–179.
21. Hutchison, C.S. (1996). The 'Rajang accretionary prism' and 'Lupar Line' problem of Borneo. *Geological Society of London Special Publication* 106.
22. Levell, B.K. (1987). The nature and significance of regional unconformities in the hydrocarbon-bearing Neogene sequence offshore West Sabah. *Bulletin of the Geological Society of Malaysia* 21: 55–90.
23. PETRONAS Malaysia Petroleum Management (2025). Exploration: overview of Malaysia's basins. petronas.com/mpm.